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PORTFOLIO 2026 BUDAPEST

— INTRODUCTION · PORTFOLIO

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Full-stack developer · TypeScript · AI-driven solutions

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About me	Projects	My books	Future learning

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01 About me

WHO I AM

> "Ever since I was a child, my dream was to work with machines and software development, and that passion has stayed with me ever since.

In **October 2022** I decided to commit to it seriously and started teaching myself to code. Starting from the basics, I worked my way up to TypeScript, which is now my main language on both the frontend and the backend.

On the frontend I work with React, Next.js and Angular, and on the backend with NestJS, Node.js and MongoDB, which makes me an **end-to-end, full-stack developer**. I have around two and a half years of commercial experience; at **BLCKS**, a Clutch Top 100 agency, I was promoted from junior to mid-level (medior).

I have worked with Redis caching and BullMQ-based multi-worker background jobs, JWT, OAuth2 and Firebase authentication, Stripe subscription billing, and Shopify/Liquid e-commerce systems. Recently I have focused strongly on **artificial intelligence**: I build solutions on top of the OpenAI and Claude APIs, voice AI systems and AI automations.

It matters to me that I can take an application all the way from planning to deployment on Vercel. In the near future I am focusing on learning **C# and .NET** as well as **SQL**, with an emphasis on the knowledge used in real-world work.

I am also preparing for the **Claude Certified Developer – Foundations (CCDV-F)** certification, and through mentoring I am deepening my knowledge of agentic coding and Claude Code, so that I can grow in the AI engineering direction.

2022
SELF-TAUGHT SINCE OCTOBER

~2.5 yrs
COMMERCIAL DEVELOPMENT EXPERIENCE

Junior → Mid
PROMOTED AT BLCKS

FRONTEND

TypeScript · React · Next.js · Angular

BACKEND

NestJS · Node.js · MongoDB · Redis · BullMQ · Stripe

AI

Claude API · OpenAI API · Voice AI · Automation

02 Projects

Champion

IN DEVELOPMENT

Football squad-building mobile game · iOS & Android

React Native · Expo · TypeScript · Next.js · MongoDB · Claude Code

PLATFORM	DATABASE	STATUS
iOS and Android	Top 5 European leagues, full squads since the 1960s	Polishing, testing and legal review

THE GAME IN SHORT

Champion is the football version of the viral NBA "82-0" squad-building game, where the goal is to assemble a team that goes through an entire league season unbeaten, 38-0. The player picks players from randomly spun teams and eras from the history of Europe's top 5 leagues, going all the way back to the 1960s. Every choice matters, because the game measures the strength of the finished squad with a season simulation.

TECHNOLOGIES

The mobile app is built with Expo and React Native in TypeScript, so a single codebase runs on both iOS and Android. The backend is powered by Next.js, and the data lives in my own MongoDB database, into which I collected every team and full squad of Europe's top 5 leagues from the 1960s to the present day.

DEVELOPMENT & DELIVERY

One of the biggest challenges was the data layer. I collected and organised all team and player data into a MongoDB database myself, so the entire dataset is in my hands and I do not depend on external APIs. The mobile app is served by a Next.js backend with the data and the endpoints the game logic needs. I build the app using agentic coding with Claude Code, which speeds up the work significantly on both the frontend and the backend.

Monetisation is based on an XP progression system and in-game coins, which can be spent on cosmetic items and convenience features. It is a deliberate design decision that coins can never buy players, so success depends solely on the player's own choices.

The project is currently in the polishing phase: I keep making smaller fixes and test the gameplay manually. Because the game uses real players' names, I am working with a lawyer on how the app can be released legally while respecting personality rights. The app will be released as an unofficial fan game, with a clear disclaimer that it is not affiliated with the clubs or the players. It will go to the App Store and Google Play once the legal questions are settled and testing is complete.

02 Projects

GridLeads

PERSONAL USE

Lead generation and outreach system · Google Maps · email and AI calls

Next.js | Vapi | MongoDB Atlas | Nodemailer | IMAP | Claude Code

DATA SOURCE	OUTREACH	TARGET MARKETS
Google Maps, custom scraper	Cold email and AI-driven phone calls	HU, US, UK, CA, AT, NL

THE SYSTEM IN SHORT

GridLeads is a lead generation and outreach system I built myself. It collects local businesses from Google Maps, scores them and then contacts them automatically. The target market can be selected by industry and town, from Hungary through the United States, the United Kingdom and Canada to Austria and the Netherlands. The system reaches out to leads both by email and with AI-driven phone calls, so the whole client acquisition process runs in one place.

TECHNOLOGIES

The Google Maps data is collected by a custom scraper, the web interface and backend are built with Next.js, and the leads are stored in a MongoDB Atlas database. Emails are sent by Nodemailer over SMTP from Google Workspace mailboxes, replies are read back via IMAP, and the automated calls are handled by the Vapi voice AI platform.

DEVELOPMENT & DELIVERY

I wrote the scraper myself, so it collects exactly the data I need and I do not depend on external services. A separate algorithm extracts email addresses for the collected businesses, so every lead can be contacted directly. The system scores leads based on, among other things, the state of their website, their reviews and their location, so the most promising businesses come first.

It was a deliberate decision to make the cold email sender part of GridLeads rather than a separate app, so collecting, scoring and outreach stay in one system. Instead of a transactional email API I use Nodemailer and IMAP, because this approach had already proven itself in an earlier automation project of mine. I contact Hungarian leads in Hungarian and international ones in English, from separate mailboxes. The sender runs 24/7: the dashboard stays open in a browser on a Kamatera server and keeps working until I stop it.

Besides email, the system also reaches out to leads with AI-driven phone calls using Vapi. I build it using agentic coding with Claude Code. GridLeads is made purely for my own use and forms the basis of my own client acquisition.

02 Projects

BlitzDeep

● LIVE · BLITZDEEP.COM

Agency landing page · the agency site behind GridLeads

Next.js | Vercel | Claude Code

WEBSITE	SERVICES	TARGET MARKET
blitzdeep.com	Websites, AI automation, social media	Primarily the US, plus Hungary

THE PROJECT IN SHORT

BlitzDeep is my own agency, offering website development, AI automation and social media marketing to local businesses. blitzdeep.com is the agency site behind GridLeads: a landing page that presents the services to the businesses the system reaches out to.

TECHNOLOGIES

The site is built with Next.js and runs on Vercel. I built it using agentic coding with Claude Code.

DEVELOPMENT & DELIVERY

The site presents the three services on separate subpages: website development, AI automation and social media marketing. Client acquisition is built on GridLeads, whose emails and calls bring interested businesses to this site. My primary target market is the United States, I also work with Hungarian clients, and I run the agency on my own.

02 Projects

OWN PROJECT · 04 / 07

KiHolMa

● LIVE · KIHOLMA.HU

Hungarian artist and concert discovery platform · public web and admin app

Next.js 15 | MongoDB Atlas | Turborepo | Auth.js v5 | MapLibre GL | Vercel | Claude Code

WEBSITE
kiholma.hu
admin.kiholma.hu

ROLES
Super admin, management, artist, fan

THE PROJECT IN SHORT

KiHolMa is a Hungarian artist and concert discovery platform that shows on a map and in a list where and when Hungarian artists perform. Concerts can be filtered by city, date, genre and price, and visitors can favourite concerts and follow artists. Anyone can submit a concert, which appears after approval, and artists and managers can manage their shows from their own accounts.

TECHNOLOGIES

The project is a pnpm and Turborepo monorepo running two Next.js 15 apps, a public website and an admin interface, with shared packages for the database models, Zod schemas and UI components. The data is stored in MongoDB Atlas with Mongoose, authentication is handled by Auth.js v5, the map is built on MapLibre GL and OpenFreeMap, emails are sent by Resend, rate limiting is provided by Upstash, hosting runs on Vercel, and quality is ensured by Vitest tests and GitHub Actions CI.

DEVELOPMENT & DELIVERY

The main element of the home page is a split-view search: the map with photo pins on the left and the concert list on the right. Venues are geocoded with Nominatim, map search uses MongoDB 2dsphere geo-indexes, and event data is stored denormalised for fast queries.

Access control is role-based: the system distinguishes the super admin, management, artist and fan roles. Publicly submitted concerts go through an approval chain, where they can be approved by the super admin, the management or the artist themselves. Moderation, full data management and featured placements are handled in the admin interface.

Users can save their favourites and follow artists in their account, synced across devices, and a daily scheduled job sends email reminders about upcoming concerts. Email features include registration confirmation, approval notifications, cancelled-concert alerts and password reset. Artists can also buy five-tier paid featured placements with duration discounts; integrating real payments is the next step.

The site also has dark mode, dynamic OG images and security headers. The two apps run as separate Vercel projects on the kiholma.hu and admin.kiholma.hu domains. I built it using agentic coding with Claude Code.

02 Projects

OWN PROJECT · 05 / 07

HEFTOR

● IN DEVELOPMENT · NEARLY DONE

Workout tracking app · iOS, Android and web

Flutter · Next.js · MongoDB · Turborepo · Apple/Google OAuth · Claude Code

PLATFORM	SURFACES
iOS, Android and web	Mobile app, web app, backend, admin

THE PROJECT IN SHORT

HEFTOR is a workout tracking app that lets users plan, log and analyse their workouts. During an active workout, sets, rest time and estimated calories burned are tracked, and afterwards detailed statistics, a calendar and a workout streak show the progress. Motivation is boosted by a rank system and a leaderboard, and a 12-week HYROX training plan is also available.

TECHNOLOGIES

The project is a Turborepo monorepo: the mobile app is built with Flutter for iOS and Android, the backend and public website run on Next.js, and there is also a Next.js web app and a separate admin interface. The data is stored in MongoDB with Mongoose, sign-in also works with Apple and Google OAuth, and the design follows my own "Monochrome Performance" design system.

DEVELOPMENT & DELIVERY

The system consists of four surfaces. Alongside the Flutter mobile app there is a Next.js backend with a landing page, blog and changelog. On mobile, the web app mirrors the phone app one to one, while on desktop it gets a wide, multi-column layout. The fourth is the admin interface. The backend serves the mobile and web clients as a REST API, with its own exercise database searchable by muscle group, body part and equipment.

Users can build their own workouts, pick ready-made ones or have the built-in generator create one. The progress page shows per-session and per-exercise details as well as 1RM and volume trend charts, and the system also flags stagnation.

The mobile app shows the weekly streak in a home-screen widget on iOS and Android, and sends a daily workout reminder via local push notification. The admin interface manages users, workouts, blog posts, the changelog, the newsletter and incoming messages. The design is deliberately minimalist: pure black OLED background, white cards, the Inter typeface and an 8-pixel grid, which the Flutter and web interfaces follow with the same design tokens.

I build it using agentic coding with Claude Code. The app is currently in development but nearly done, and the premium subscription plan is still being designed.

02 Projects

OWN PROJECT · 06 / 07

FixLibrary

IN DEVELOPMENT

Mac desktop app for cleaning up a DJ music library · Rekordbox integration

Electron · TypeScript · React · Python · SQLite · Demucs · Essentia · Claude Code

PLATFORM	LIBRARY	INTEGRATION
macOS (Apple Silicon)	~3 000 tracks, mostly MP3	Rekordbox XML

THE PROJECT IN SHORT

FixLibrary is a Mac desktop app I am building for my own use to clean up my DJ music library, so that every track works well in Rekordbox and on Pioneer players. It fixes filenames and tags, finds cover art, organises tracks into folders by style, calculates energy levels, places hot cues automatically, recommends and generates playlists, and normalises loudness. It hands the result over through Rekordbox XML, so it never touches Rekordbox's own database.

TECHNOLOGIES

The app is built with Electron, React and TypeScript, and the data is stored in SQLite with Drizzle ORM. Audio analysis is done by a Python sidecar: Demucs handles stem separation, Essentia the audio analysis, pyloudnorm the loudness normalisation and yt-dlp the downloads. For AI-driven decisions it uses the Claude CLI, without a separate API key.

DEVELOPMENT & DELIVERY

Development started with detailed documentation, in which I specified the behaviour feature by feature, and continued with a 45-task development plan. Filenames get a consistent format: artist, title and mix, plus Camelot and musical key, BPM and energy level, so the library is easy to scan from the filenames alone.

For hot cues the app uses eight fixed slots (such as bassline, vocal, drop and breakdown), which it detects from the bass and vocal stems separated by Demucs. Analysis of the full library runs in the background and can be paused and resumed. Getting everything into Rekordbox is guided by a step-by-step wizard, which also checks the export to confirm that every track and hot cue landed in the right place.

The app can either modify files in place or create a separate, clean copy of the library. It receives updates through GitHub Releases and installs as a single .dmg file. I build it using agentic coding with Claude Code.

02 Projects

OWN PROJECT · 07 / 07

Melodia

PERSONAL USE

Job search outreach dashboard · research, writing, sending and replies in one place

Next.js 16 · MongoDB Atlas · React 19 · Nodemailer · ImapFlow · Claude CLI · OpenAI

USE	DATABASE	AI
My own job search	~290 000 companies	Claude CLI, optionally OpenAI

THE PROJECT IN SHORT

Melodia is an outreach dashboard I built for my own job search. It runs the whole process in one place: researching companies, writing personalised emails, sending them and handling incoming replies. The database holds around 290 000 companies, for which the system finds email addresses and contacts using web search and AI. The list is sorted by a fit score, so I always start with the most promising companies.

TECHNOLOGIES

The dashboard is built with Next.js 16 and React 19, and the data is stored in MongoDB Atlas. Emails are sent by Nodemailer from Gmail accounts, and incoming replies are synced with ImapFlow. AI tasks (email generation, reply classification, interview briefs, email search) are handled by the local Claude CLI, with OpenAI available only as an optional search engine.

DEVELOPMENT & DELIVERY

Data comes in several ways: from PDF sources, Hunter CSV exports, JSON imports or just a company name, which the system adds and researches. Imports filter out duplicates by key, company name alias and domain. The fit score is calculated by the database from factors such as stack match, location, company size and whether an HR contact was found.

Emails go out during working hours in the recipient's time zone, with a daily cap and a gradual warm-up for new accounts. A pre-send check filters out missing or incomplete emails, generic mailboxes and domains that have bounced before. Gmail sync runs every 30 minutes and writes replies and bounces back to the contacts.

Incoming replies are classified by AI (interview, question, rejection, later), summarised in one sentence, and an editable draft reply is prepared in the language of the reply that never commits to anything on my behalf. The system also prepares follow-up emails in the same thread, and for interviews it writes a preparation brief about the company. By using custom system prompts and a restricted tool set, I cut the Claude CLI's token usage from around 70 thousand to around 4 thousand tokens per task.

02 Projects

Client projects · BLCKS

AGENCY CLIENT WORK

CLIENT PROJECT 01 / 04

Flinkit

AI-powered interactive game platform · team building, events, tourism

#2 PRODUCT HUNT

TECHNOLOGIES	MY ROLE	CLIENT	RESULT
Angular, NestJS, MongoDB, Azure	Full-stack developer	BLCKS client project	Product Hunt #2

Flinkit is an AI-powered interactive game platform that lets companies, event organisers, museums and tourism providers create team-building games, quizzes and location-based challenges. Participants join from a mobile browser without installing an app, while organisers create and customise the games in an admin interface, even with the help of AI. The platform is also available as a white-label solution, so partners can offer it under their own brand.

TECHNOLOGIES

The platform's frontend is built with Angular and its backend with NestJS, the data is stored in MongoDB, and the application runs on Azure.

DEVELOPMENT & DELIVERY

I worked on Flinkit at BLCKS as a full-stack developer. My job was building new features end to end: I analysed the clients' requirements and then implemented the requested features on both the frontend and the backend. The system's complexity comes from handling dynamically generated games and tasks, GPS and map integrations, AI APIs, and user and admin systems all at once. Since it also has to serve large events at the same time, scalability is a key concern. The product reached #2 on Product Hunt.

02 Projects

Client projects · BLCKS

AGENCY CLIENT WORK

CLIENT PROJECT 02 / 04

Wingman

Travel mobile app · Flutter and NestJS

100 000+
DOWNLOADS

TECHNOLOGIES	MY ROLE	CLIENT	RESULT
Flutter, NestJS	Restructure and product direction	BLCKS client project	Monthly ~50 → ~1 000 downloads

Wingman is a travel mobile app that has reached more than 100 000 downloads. My work on the project was the complete restructuring of the app and setting its new product direction.

TECHNOLOGIES

The mobile app is built with Flutter, and the backend is powered by NestJS.

DEVELOPMENT & DELIVERY

The overhaul started on the product side: I cut the features that brought users little value and added ones that were trending in other apps. Along with this, the app's design direction changed as well. On the technical side, I restructured both the backend and the frontend for better performance: I introduced caching and simplified the user flow. After the overhaul, monthly downloads grew from around 50 to around 1 000.

02 Projects

Client projects · BLCKS

AGENCY CLIENT WORK

CLIENT PROJECT 03 / 04

Opentrends

Local SEO analysis and improvement platform · automated cold outreach

B2B PLATFORM

TECHNOLOGIES	MY ROLE	CLIENT	AREA
Angular, NestJS, MongoDB, Vapi, Resend	Full-stack developer	BLCKS client project	Local SEO and client acquisition

Opentrends is a B2B platform that analyses and improves the online visibility of local businesses (local SEO). The system tracks businesses' Google Search and Google Maps rankings, analyses nearby competitors, monitors online reviews and gives AI-driven recommendations for improving rankings. It also automates client acquisition: it reaches out to businesses selected by given criteria with automated cold emails and phone calls.

TECHNOLOGIES

The platform's frontend is built with Angular and its backend with NestJS, and the data is stored in MongoDB. The automated phone calls are handled by the Vapi voice AI platform, and the emails are sent by Resend.

DEVELOPMENT & DELIVERY

I worked on Opentrends at BLCKS as a full-stack developer and took part in developing every part of the platform, on both the frontend and the backend. This included the local SEO analysis, meaning tracking search and Maps rankings, competitor analysis and review monitoring, as well as the dashboard where results and recommendations appear in one view. The other main part of the platform is automated client acquisition: the system reaches out to the selected businesses with cold emails sent via Resend and Vapi-based AI calls.

02 Projects

Client projects · BLCKS

AGENCY CLIENT WORK

CLIENT PROJECT 04 / 04

Apesyto

Webshop and partner programme · own-brand products, community sales

E-commerce
WEBSHOP

TECHNOLOGIES	MY ROLE	CLIENT	AREA
NestJS, BullMQ, Redis, Angular, MongoDB, WooCommerce	Partner programme development	BLCKS client project	E-commerce and partner programme

Apesyto is an e-commerce brand that sells its own-brand products through a webshop with a community sales model. Alongside the webshop runs a partner programme offering partners 12 different payout and benefit options, including gold investment and an invitation-based, multi-level (MLM) partner network.

TECHNOLOGIES

The platform's frontend is built with Angular and its backend with NestJS, the data is stored in MongoDB, and the webshop runs on WooCommerce. Commission calculation is powered by BullMQ-based background jobs and Redis caching.

DEVELOPMENT & DELIVERY

I worked on Apesyto at BLCKS and built several key parts of the partner programme. My work included implementing five of the 12 benefit types, the gold investment option, and the invitation system through which partners can invite new members to the network. These features are closely linked: the partner network is built from invitations, and benefits are paid to members according to the programme's terms. To calculate commissions, I used BullMQ-based background jobs running in parallel across multiple workers, together with Redis caching, so that partner payouts are produced accurately and quickly.

03 My books

MY WRITING

I wrote the following Hungarian-language technical books with the help of Claude. I learn a new subject by building a book out of it chapter by chapter, keeping the technical terms in English and explaining them in Hungarian.

6	3 500+	3
BOOKS	PAGES IN TOTAL	FINISHED BOOKS

IN PROGRESS · ~1 750 PAGES · 14/30 CHAPTERS DONE

Medior to Senior

A comprehensive professional handbook for mid-level full-stack TypeScript developers who want to reach senior level. The finished chapters cover automated testing, SQL and PostgreSQL, Docker, web security, performance optimisation, caching, real-time communication, and Python for TypeScript developers. The examples come from code that was actually run and from measured results.

IN PROGRESS · 450+ PAGES · 24 CHAPTERS · 13 PARTS PLANNED

Claude Code & Agentic Coding

A beginner-friendly book that goes from the basics of LLMs and agentic coding to building a Claude Code harness, context engineering and memory management. The content is based on the official Claude Code documentation and Anthropic's engineering blog, with many original diagrams and exercises.

DONE · 443 PAGES · 21 CHAPTERS

C# & .NET

A complete C# and .NET course from scratch: from the language basics through ASP.NET Core Web API, EF Core and testing to Clean Architecture and CI/CD. It closes by building a complete library loan management system, with JavaScript and NestJS parallels throughout.

DONE · 418 PAGES · 24 CHAPTERS AND APPENDIX

SQL

A T-SQL and SQL Server book for complete beginners, from the Docker setup through JOINs, indexes and window functions to .NET integration and performance diagnostics, with a PostgreSQL comparison. It closes with a capstone project that builds a ticketing system and an appendix of interview questions.

IN PROGRESS · 386 PAGES · 12 PARTS PLANNED

AI Engineer

A book for complete beginners based on the roadmap.sh AI Engineer roadmap, from the core concepts (tokens, embeddings, RAG, agents) through pre-trained models to the OpenAI Platform API and prompt writing. Technical terms stay in English but are explained in detail in Hungarian.

DONE · 151 PAGES · 30 TOPICS

The Next Thirty Steps for a Mid-Level Developer

A learning plan of 30 professional topics, from automated testing and SQL to RAG architecture. Each topic comes with a detailed description and a ready-to-paste prompt; these were used to produce the chapters of Medior to Senior.

04 Future learning

WHAT'S NEXT

In the coming period I am focusing on these areas, always with an emphasis on the knowledge used in real-world work. I learn new subjects by writing a book about them as I go, and I immediately try out what I learn in my own projects.

01 C# and .NET

IN PROGRESS

Alongside my TypeScript and NestJS experience, I am learning a second backend ecosystem that is widely used in enterprise environments. The focus is on the knowledge used in real-world work: beyond the language basics, ASP.NET Core Web API, EF Core, testing and Clean Architecture.

HOW: Based on my own C# & .NET book, by building a complete library loan management system.

02 SQL and relational databases

IN PROGRESS

So far I have mostly worked with MongoDB, so I am learning relational databases from the ground up: from queries and JOINs through indexes, transactions and window functions to performance diagnostics. The main focus is T-SQL and SQL Server, with a PostgreSQL comparison.

HOW: Based on my own SQL book, through practice databases and a capstone project.

03 Claude Certified Developer – Foundations

CERTIFICATION

In September 2026 I started preparing for Anthropic's official developer certification, the CCDV-F. The goal is to have my hands-on experience building with Claude officially certified.

HOW: Through the Anthropic Partner Academy learning path.

04 Agentic coding and Claude Code

MENTORING

Through mentoring I am deepening my agentic coding skills: the next step is designing my own Claude Code harness, that is, which skills, hooks, commands and scheduled tasks an effective workflow needs. This will be followed by a makerspace programme for developers.

HOW: Through regular mentoring sessions and by writing my own Claude Code & Agentic Coding book.

05 Towards AI engineering

LONG TERM

In the longer term I want to grow in the AI engineering direction, so I am following the roadmap.sh AI Engineer roadmap: from how LLMs and embeddings work through RAG and agents to AI-driven product development.

HOW: By writing my own AI Engineer book and applying what I learn in my own projects.

— CONTACT

Thank you for your time!

If you would like to talk about a project or working together, please reach out through any of my contact details.

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